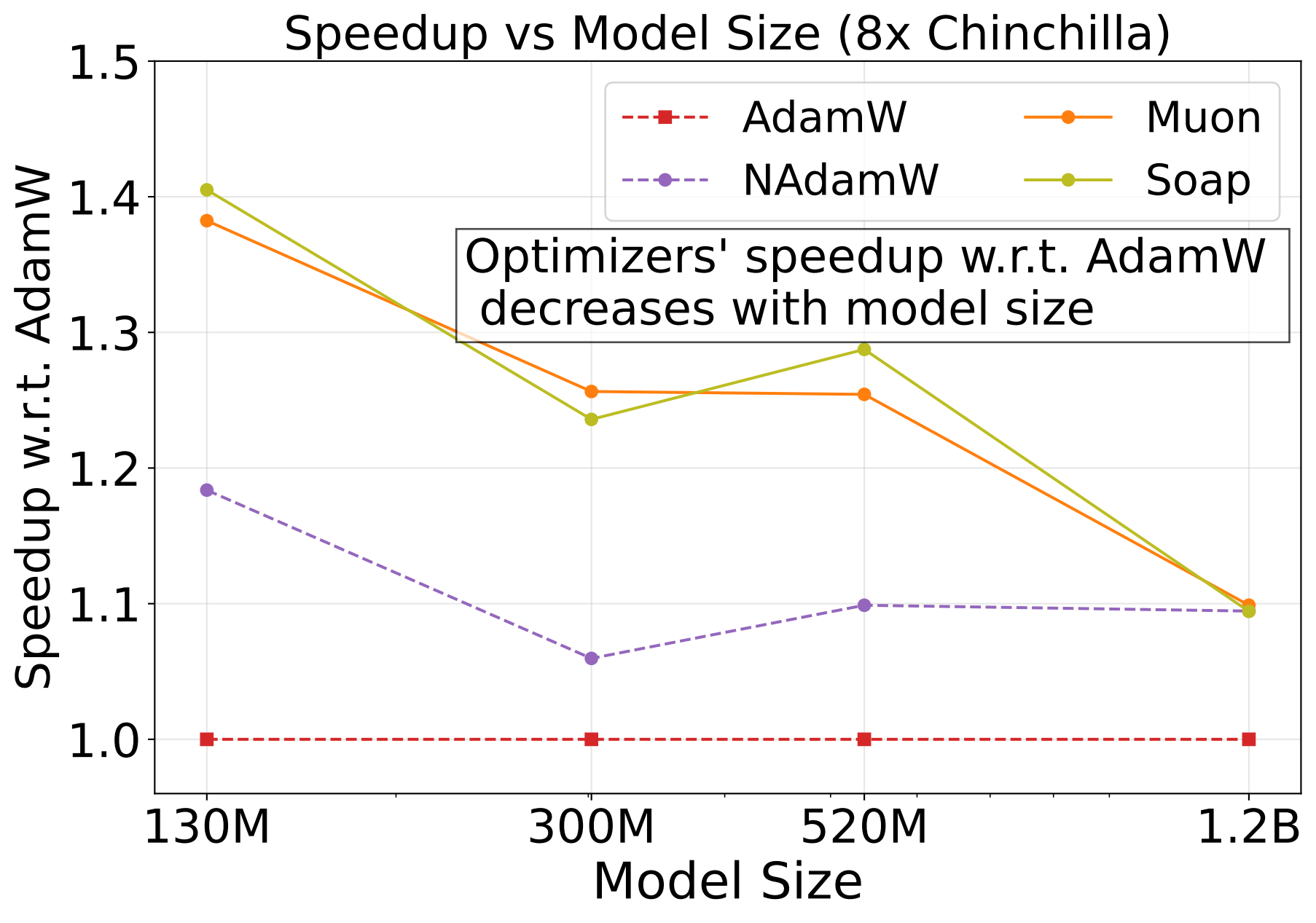
Speedup vs Model Size (8x Chinchilla)
AdamW
Muon
NAdamW
Soap
Optimizers' speedup w.r.t. AdamW
decreases with model size
Speedup w.r.t. AdamW
1.5
1.4
1.3
1.2
1.1
1.0
130M
300M
520M
1.2B
Model Size